An exact two-mode characterization of Chambolle–Pock five-cycles

Abstract

We give an exact second-order-cone feasibility characterization of the stepsize products admitting a nonconstant five-periodic orbit for the literal primal-first Chambolle–Pock recurrence. The necessity argument applies to arbitrary finite-dimensional real Hilbert spaces, arbitrary linear coupling, and arbitrary proper closed convex functions. Conversely, every feasible point yields explicit indicator functions in dimension eight, a least-period-five orbit, an actual initialization, and a saddle point. All eight cyclic interpolation inequalities are retained; no rank condition or unverified interpolation constraint remains. Exact certificates exclude products at most $31/10$ when $\theta = 1/8$ and at most $17/10$ when $\theta = 1$, while explicit two-mode examples exist at $25/8$ and $7/4$, respectively. The certificates use only rational arithmetic in $\mathbb{Q}(\sqrt{5})$. These are certified brackets, not sharp scalar thresholds or a global convergence result. No novelty claim is made.

Keywords: Chambolle–Pock iteration; period-five orbits; second-order-cone feasibility; Fourier modes; convex interpolation; exact certificates

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1 The recurrence and the question

Let X, Y be finite-dimensional real Hilbert spaces, $L : X \to Y$ linear, and f, h proper closed convex extended-real-valued functions on X, Y, respectively. For $a > 0$ define

$$\operatorname{prox}_{af}(z) = \arg\min_x \{f(x) + \|x - z\|^2/(2a)\}.$$

For $\theta \geq 0$, $\tau, \sigma > 0$, consider exactly the paired-state map

$$\begin{aligned} x_{i+1} &= \operatorname{prox}_{\tau f}(x_i - \tau L^* y_i), \\ y_{i+1} &= \operatorname{prox}_{\sigma h}\big(y_i + \sigma L((1+\theta)x_{i+1} - \theta x_i)\big). \end{aligned} \tag{1}$$

The dual step uses the newly computed primal state; no extra extrapolated state is initialized. Write $r = \tau\sigma\|L\|^2 > 0$. Chambolle and Pock [1] provide the primal–dual framework; the order and parameter scope here are specified by (1), not attributed to a convergence theorem in that reference.

2 The exact finite conic system

Put $\phi_k = 2\pi k/5$, $C_k = \cos\phi_k$, $s_k = \sin\phi_k > 0$, for $k = 1, 2$. In particular,

$$C_1 = (\sqrt{5}-1)/4, \quad C_2 = -(\sqrt{5}+1)/4, \quad s_1^2 = (5+\sqrt{5})/8, \quad s_2^2 = (5-\sqrt{5})/8.$$

For $j = 1, 2, 3, 4$ define the exact coefficients

$$\begin{aligned} D_{kj} &= \cos((j+1)\phi_k) - \cos\phi_k - \cos(j\phi_k) + 1, \\ Z_{kj} &= e^{\mathrm{i}\phi_k}(e^{\mathrm{i}j\phi_k} - 1), \\ W_{kj} &= ((1+\theta) - \theta e^{\mathrm{i}\phi_k})(e^{\mathrm{i}j\phi_k} - 1). \end{aligned} \tag{2}$$

The unknowns are the eight real numbers A_k, B_k, c_k, d_k, $k = 1, 2$. Impose

$$A_k, B_k \geq 0, \qquad c_k^2 + d_k^2 \leq A_k B_k, \qquad A_1 + A_2 = 1, \tag{3}$$

and all eight linear inequalities

$$\begin{aligned} P_j &= \sum_{k=1}^{2} \left(\frac{D_{kj}}{r} A_k - (\operatorname{Re} Z_{kj}) c_k - (\operatorname{Im} Z_{kj}) d_k \right) \leq 0, \\ H_j &= \sum_{k=1}^{2} (D_{kj} B_k + (\operatorname{Re} W_{kj}) c_k - (\operatorname{Im} W_{kj}) d_k) \leq 0, \quad j = 1, 2, 3, 4. \end{aligned} \tag{4}$$

For fixed (θ, r) this is a second-order-cone (SOC) system: its quadratic constraints are equivalently

$$\|(2c_k, 2d_k, A_k - B_k)\|_2 \leq A_k + B_k.$$

Equivalently, each Hermitian mode Gram matrix $\left(\begin{smallmatrix} A_k & c_k - \mathrm{i}d_k \\ c_k + \mathrm{i}d_k & B_k \end{smallmatrix}\right)$ is positive semidefinite. This equivalent notation does not impose a rank restriction.

Theorem 1. *For each $\theta \geq 0$ and $r > 0$, the following are equivalent:*

1. *Some map* (1) *with actual product r has an orbit of least period five.*
2. *Such a map exists with a saddle point as well as an orbit of least period five.*
3. *The complete system* (3)–(4) *is feasible.*

Every feasible point gives a realization with $X = Y = \mathbb{R}^8$, $L = I$, $\tau = r$, $\sigma = 1$, and indicators of explicitly specified convex hulls. The saddle pair is $(0, 0)$.

The conic system is an exact implicit product-region characterization, not an SDP relaxation or a planar regular-polygon ansatz. The proof below establishes the absence of missing function-value constraints and additional rank conditions.

3 Necessity for arbitrary convex problems

Index a five-periodic orbit modulo five. Every state is a proximal output, so the function values there are finite. Proximal optimality yields the actual subgradients

$$\begin{aligned} a_i &= (x_{i-1} - x_i)/\tau - L^* y_{i-1} \in \partial f(x_i), \\ b_i &= (y_{i-1} - y_i)/\sigma + L((1+\theta)x_i - \theta x_{i-1}) \in \partial h(y_i). \end{aligned} \tag{5}$$

For each $j = 1, 2, 3, 4$, sum the five subgradient inequalities. Since $i \mapsto i + j$ permutes the phases, the function values telescope and give

$$\sum_i \langle a_i, x_{i+j} - x_i \rangle \le 0, \qquad \sum_i \langle b_i, y_{i+j} - y_i \rangle \le 0. \tag{6}$$

This uses actual convex interpolation, not only pairwise monotonicity.

Let $\bar{x} = \frac{1}{5}\sum_i x_i$, $\bar{y} = \frac{1}{5}\sum_i y_i$, and set

$$U_i = (x_i - \bar{x})/\sqrt{\tau}, \quad V_i = (y_i - \bar{y})/\sqrt{\sigma}, \quad K = \sqrt{\tau\sigma}\, L.$$

Equip temporal direct sums with $\langle U, V \rangle_{\mathrm{av}} = \frac{1}{5}\sum_i \langle U_i, V_i \rangle$ when the spatial spaces agree. Let $(FU)_i = U_{i+1}$ and $S = F^{-1}$. The constant terms vanish when paired with differences, and (6), divided by five, becomes

$$\begin{aligned} \langle (S - I)U - K^* S V, (F^j - I)U \rangle_{\mathrm{av}} \le 0, \\ \langle (S - I)V + K((1+\theta)I - \theta S)U, (F^j - I)V \rangle_{\mathrm{av}} \le 0. \end{aligned} \tag{7}$$

Here K is lifted diagonally to the temporal spaces and has norm $\sqrt{r}$.

The centered temporal space is the orthogonal direct sum of its two real frequency subspaces. On subspace k,

$$F = C_k I + s_k J_k, \qquad J_k^* = -J_k, \qquad J_k^2 = -I.$$

For completeness, the orthogonal mode projector is $\Pi_k = \frac{2}{5}\sum_{m=0}^{4} \cos(m\phi_k) F^m$. The identities follow by summing the fifth roots of unity; on centered sequences $\Pi_1 + \Pi_2 = I$. The diagonal lift K commutes with shifts and intertwines the primal and dual copies of J_k. Write $U_k = \Pi_k U$, $V_k = \Pi_k V$ and, for the moment, define unnormalized quantities

$$A_k = r\|U_k\|_{\mathrm{av}}^2, \quad B_k = \|V_k\|_{\mathrm{av}}^2, \quad c_k = \langle K U_k, V_k \rangle_{\mathrm{av}}, \quad d_k = \langle K J_k U_k, V_k \rangle_{\mathrm{av}}. \tag{8}$$

3.1 Gram bound, signs, and normalization

Because $KJ_kU_k = J_kKU_k$, skew-adjointness gives

$$\langle KU_k, KJ_kU_k\rangle_{\mathrm{av}} = 0, \qquad \|KJ_kU_k\|_{\mathrm{av}} = \|KU_k\|_{\mathrm{av}}.$$

Bessel's inequality, with the zero-vector case treated directly, therefore gives

$$c_k^2 + d_k^2 \le \|KU_k\|_{\mathrm{av}}^2 \|V_k\|_{\mathrm{av}}^2 \le r\|U_k\|_{\mathrm{av}}^2 \|V_k\|_{\mathrm{av}}^2 = A_kB_k. \tag{9}$$

This is where the actual coupling norm enters; no isometry assumption on the original L is made.

The self terms in (7) have coefficient

$$\mathrm{Re}\big[(e^{\mathrm{i}\phi_k} - 1)(e^{\mathrm{i}j\phi_k} - 1)\big] = D_{kj}.$$

For the primal cross term, moving S to the other slot gives

$$-\langle SV_k, K(F^j - I)U_k\rangle_{\mathrm{av}} = -\langle V_k, KF(F^j - I)U_k\rangle_{\mathrm{av}} = -(\mathrm{Re}\, Z_{kj})c_k - (\mathrm{Im}\, Z_{kj})d_k.$$

For the dual cross term the corresponding product in the first slot is

$$(F^{-j} - I)((1+\theta)I - \theta S) = \overline{W_{kj}}$$

on this mode. Its contribution is consequently $(\mathrm{Re}\, W_{kj})c_k - (\mathrm{Im}\, W_{kj})d_k$. Orthogonality of the modes removes cross-mode terms. Substitution of (8) proves all of (4), including their displayed orientations.

It remains to justify normalization rather than assume primal variation. If $U = 0$, take $j = 4$ in the dual inequality of (7); since $F^4 = S$, it reads

$$\|(S - I)V\|_{\mathrm{av}}^2 \le 0.$$

Thus V is temporally constant and, being centered, vanishes. A nonconstant paired orbit must therefore have $q = A_1 + A_2 > 0$. Divide all eight variables in (8) by q. The linear inequalities scale by $1/q$ and both sides of the Gram inequality by $1/q^2$, establishing (3)–(4). Necessity is proved without any saddle-point assumption.

A nonconstant five-periodic orbit has least period five: the least period of a periodic sequence divides every period, and five is prime. The realization below also proves distinctness directly, without relying only on this divisibility observation.

4 Every feasible point is a genuine orbit

Take any feasible point. Decompose $\mathbb{R}^8$ as four mutually orthogonal coordinate planes $E_{k,\ell}$, $k, \ell = 1, 2$. Let T rotate each $E_{k,\ell}$ counterclockwise by ϕ_k, using the usual orientation $J(u, v) = (-v, u)$. Then

$$T^5 = I, \qquad \sum_{i=0}^{4} T^i = 0. \tag{10}$$

For each k with $A_k > 0$, specify the components of x_0, y_0 by

$$\begin{aligned} x_{0,k} &= (\sqrt{A_k}, 0) \text{ in } E_{k,1}, \quad 0 \text{ in } E_{k,2}, \\ y_{0,k} &= (c_k/\sqrt{A_k}, d_k/\sqrt{A_k}) \text{ in } E_{k,1}, \\ &\quad (\sqrt{B_k - (c_k^2 + d_k^2)/A_k}, 0) \text{ in } E_{k,2}. \end{aligned} \tag{11}$$

The radicand is nonnegative by (3). If $A_k = 0$, the cone forces $c_k = d_k = 0$; define $x_{0,k} = 0$ and put $y_{0,k} = (\sqrt{B_k}, 0)$ in $E_{k,2}$. This includes $B_k = 0$ and all singular Gram cases. In every case,

$$\|x_{0,k}\|^2 = A_k, \quad \|y_{0,k}\|^2 = B_k, \quad \langle x_{0,k}, y_{0,k} \rangle = c_k, \quad \langle Jx_{0,k}, y_{0,k} \rangle = d_k.$$

Sum over k and set

$$\begin{aligned} x_i = T^i x_0, \quad y_i = T^i y_0, \quad C = \operatorname{conv}\{x_0, \ldots, x_4\}, \quad D = \operatorname{conv}\{y_0, \ldots, y_4\}, \\ f = \delta_C, \quad h = \delta_D, \quad L = I_{\mathbb{R}^8}, \quad \tau = r, \quad \sigma = 1. \end{aligned} \tag{12}$$

Here δ_C is zero on C and $+\infty$ elsewhere. Both sets are nonempty compact convex polytopes, so the functions are proper closed convex. Crucially $\|L\| = 1$ on the full ambient space, even if the points span a smaller subspace; the actual product is exactly r.

Define, with indices modulo five,

$$\begin{aligned} a_i &= (x_{i-1} - x_i)/r - y_{i-1}, \\ b_i &= y_{i-1} - y_i + (1 + \theta)x_i - \theta x_{i-1}. \end{aligned} \tag{13}$$

The mode inner products above and the rotation orientation give, directly,

$$\langle a_i, x_{i+j} - x_i \rangle = P_j \le 0, \qquad \langle b_i, y_{i+j} - y_i \rangle = H_j \le 0 \quad (0 \le i \le 4,\ 1 \le j \le 4). \tag{14}$$

One can compute at $i = 0$ using (2); orthogonality of T^i then proves every other phase. These are all twenty off-diagonal vertex inequalities for each polytope, hence all forty required inequalities, not only adjacent-vertex tests.

4.1 Proximal transitions, least period, and the saddle

Every point of C is a convex combination of its displayed generators, so (14) implies $\langle a_i, z - x_i\rangle \le 0$ for all $z \in C$. Consequently $a_i \in N_C(x_i) = \partial\delta_C(x_i)$; likewise $b_i \in N_D(y_i)$. Rewriting (13) gives

$$x_{i-1} - ry_{i-1} - x_i \in rN_C(x_i),$$

$$y_{i-1} + (1+\theta)x_i - \theta x_{i-1} - y_i \in N_D(y_i).$$

These are exactly the two projection optimality conditions for (1). Projections onto nonempty closed convex sets are single-valued: strict convexity of the squared-distance objective gives uniqueness. Thus the literal initial pair (x_0, y_0) follows the stated sequence and returns after five steps. There is no prescribed hidden state or set-valued selection.

Normalization gives $\|x_0\|^2 = A_1 + A_2 = 1$. For $j = 1, 2, 3, 4$, neither kj is divisible by five for $k = 1, 2$. Hence $T^j - I$ is invertible on each of the four planes. It follows that $x_j \neq x_0$, and applying T^i shows that all five primal phases are distinct. The paired orbit has least period five.

By (10), both point sets have centroid zero, so $0 \in C \cap D$. The saddle expression is

$$\mathcal{L}(x, y) = \delta_C(x) + \langle x, y\rangle - \delta_D(y).$$

For $x \in C, y \in D$ one has $\mathcal{L}(0, y) = \mathcal{L}(0, 0) = \mathcal{L}(x, 0) = 0$; outside these domains the usual extended-real saddle inequalities remain valid. Equivalently, $0 \in N_C(0)$ and $0 \in N_D(0)$ prove the primal–dual optimality inclusions at $(0, 0)$.

This proves sufficiency with a saddle point. Saddle-restricted existence trivially implies unrestricted existence, completing Theorem 1.

4.2 Why no extra rank or interpolation restriction is needed

The cone permits full-rank or singular mode Gram matrices. Formula (11) realizes every one, including the $A_k = 0$ boundary, using the additional plane for its residual norm. Nothing requires a zero determinant. The indicators provide constant finite values on the orbit, and (14) proves all vertex normal conditions needed for those functions. Thus there are no omitted function values to interpolate and no extra rank constraints to enforce.

The original orbit in the necessity direction was not assumed rotationally symmetric. The exact temporal decomposition accounts for both frequencies of every centered five-phase sequence, in arbitrary spatial dimension. Only the constructed witness is rotationally symmetric in the larger direct sum. It may use both frequencies and the residual planes, so it is generally not a regular pentagon or pentagram in a single spatial plane.

5 Certified exclusions and two-mode examples

Corollary 5.1. *At $\theta = 1/8$, no nonconstant five-periodic orbit exists for $0 \le r \le 31/10$, and a least-period-five orbit with a saddle point exists at $r = 25/8$. At $\theta = 1$, no nonconstant five-periodic orbit exists for $0 \le r \le 17/10$, and a least-period-five orbit with a saddle point exists at $r = 7/4$.*

All exclusions range over the full convex class, without a saddle assumption. The case $r = 0$ is included in this corollary separately from Theorem 1. If $\rho_5(\theta)$ denotes the infimum of admissible positive products, this proves only the certified brackets

$$31/10 \le \rho_5(1/8) \le 25/8, \qquad 17/10 \le \rho_5(1) \le 7/4.$$

The lower endpoints themselves are excluded. No assertion of a sharp scalar infimum, attainment of that infimum, or feasibility of every product inside either bracket is made.

At $r = 0$, positivity of the stepsizes forces $L = 0$. Each component is then an uncoupled proximal iteration. Taking the preceding iterate as the target in its subgradient inequality and summing around the cycle yields $\sum_i \|x_{i-1} - x_i\|^2/\tau \le 0$, and likewise for y. Both sequences are constant. This proves the zero-product exclusion separately.

5.1 Exact feasible data

Write $d_k = s_k e_k$. Every entry in each of the following arrays is divided by 10^6:

$\theta = 1/8,\ r = 25/8$	$k = 1$	$k = 2$
A_k	77976	922024
B_k	20760	358860
c_k	10741	554773
e_k	−40280	−255973

(15)

$\theta = 1,\ r = 7/4$	$k = 1$	$k = 2$
A_k	309920	690080
B_k	308005	1236296
c_k	201304	912361
e_k	−244008	−224224

(16)

In each case $A_k, B_k > 0$ and $A_1 + A_2 = 1$. The two determinants $A_k B_k - c_k^2 - s_k^2 e_k^2$ are strictly positive and each of the eight expressions P_j, H_j is strictly negative. Exact expanded values are provided below. Substitution into (11) and (12) is a literal radical-coordinate construction of the functions and starting pair, rather than an existence statement based on numerical optimization. Both primal modes are nonzero. All forty off-diagonal inequalities in (14) are strict for these examples.

5.2 Universal exclusion by nonnegative weights

At $(\theta, r_0) = (1/8, 31/10)$ use

$$\lambda = (407, 0, 133, 0), \qquad \mu = (332, 2, 126, 0).$$

At $(\theta, r_0) = (1, 17/10)$ use

$$\lambda = (308, 0, 401, 0), \qquad \mu = (74, 13, 204, 0).$$

Multiply the inequalities $P_j \le 0, H_j \le 0$ by λ_j, μ_j and sum. The result is

$$\sum_{k=1}^{2}(a_k A_k + b_k B_k + g_k c_k + h_k d_k) \le 0, \tag{17}$$

where the coefficients, with their full signs, are

$$\begin{aligned} a_k &= r^{-1}\sum_{j=1}^{4}\lambda_j D_{kj}, \qquad b_k = \sum_{j=1}^{4}\mu_j D_{kj}, \\ g_k &= \sum_{j=1}^{4}(-\lambda_j \operatorname{Re} Z_{kj} + \mu_j \operatorname{Re} W_{kj}), \\ h_k &= \sum_{j=1}^{4}(-\lambda_j \operatorname{Im} Z_{kj} - \mu_j \operatorname{Im} W_{kj}). \end{aligned} \tag{18}$$

Let $\varepsilon_1 = -1$, $\varepsilon_2 = 1$, and $z = \sqrt{5}$. Exact expansion at $r = r_0$ gives

$$\begin{aligned} \theta = 1/8: \quad a_k &= (6750 + \varepsilon_k 2720z)/31, \\ b_k &= 575 + \varepsilon_k 218z, \\ 4a_k b_k - g_k^2 - h_k^2 &= (3980285 + \varepsilon_k 1556275z)/992; \end{aligned} \tag{19}$$

$$\begin{aligned} \theta = 1: \quad a_k &= (17725 + \varepsilon_k 2615z)/34, \\ b_k &= (1455 + \varepsilon_k 31z)/4, \\ 4a_k b_k - g_k^2 - h_k^2 &= (479895 + \varepsilon_k 42267z)/34. \end{aligned} \tag{20}$$

Each displayed quantity is strictly positive. These signs follow by rational squaring, as detailed in the exact-arithmetic section. For $0 < r \le r_0$, only a_k changes: it is multiplied by $r_0/r \ge 1$. Therefore all three strict inequalities remain true over the entire lower-product range.

By (3) and Cauchy–Schwarz,

$$a_k A_k + b_k B_k + g_k c_k + h_k d_k \ge a_k A_k + b_k B_k - \sqrt{g_k^2 + h_k^2}\sqrt{A_k B_k}.$$

If $A_k B_k > 0$, the right side is strictly positive, since $\sqrt{g_k^2 + h_k^2} < 2\sqrt{a_k b_k}$ and $a_k A_k + b_k B_k \ge 2\sqrt{a_k b_k A_k B_k}$. If exactly one of A_k, B_k is positive, positivity follows from $a_k, b_k > 0$; if both vanish, the contribution is zero. Since $A_1 + A_2 = 1$, the sum in (17) is strictly positive, a contradiction. This proves both exclusions for positive products. The zero-product case proved above completes Corollary 5.1.

6 Exact arithmetic and reproducibility

All certificate checks reduce to $\mathbb{Q}(\sqrt{5})$. To avoid treating trigonometric or floating-point output as a sign proof, let T_n, U_n denote the Chebyshev polynomials, defined by

$$T_0 = 1,\ T_1 = C, \quad U_0 = 1,\ U_1 = 2C, \qquad Q_{n+1} = 2CQ_n - Q_{n-1}$$

for either family. Set $q_0 = 0$ and $q_n = U_{n-1}(C_k)$ for $n \geq 1$. Thus $\cos(n\phi_k) = T_n(C_k)$ and $\sin(n\phi_k) = s_k q_n$. For each mode the coefficients in (2) can be evaluated as

$$\begin{aligned} D_{kj} &= T_{j+1}(C_k) - C_k - T_j(C_k) + 1, \\ \operatorname{Re} Z_{kj} &= T_{j+1}(C_k) - C_k, & \operatorname{Im} Z_{kj} &= s_k(q_{j+1} - 1), \\ \operatorname{Re} W_{kj} &= (1+\theta)(T_j(C_k) - 1) - \theta \operatorname{Re} Z_{kj}, \\ \operatorname{Im} W_{kj} &= s_k((1+\theta)q_j - \theta(q_{j+1} - 1)). \end{aligned}$$

With $d_k = s_k e_k$, every primal normal expression, Gram determinant, and dual determinant lies in $\mathbb{Q}(\sqrt{5})$. The verifier uses precisely these formulas.

For $u = a + b\sqrt{5}$, $a, b \in \mathbb{Q}$, the sign is determined exactly as follows. If $a = 0$ or $b = 0$, use the sign of the other coefficient. If both have the same sign, use that sign. If they have opposite signs, then

$$\operatorname{sign}(u) = \operatorname{sign}(a)\ \operatorname{sign}(a^2 - 5b^2).$$

Indeed $a - b\sqrt{5}$ has the sign of a and $u(a - b\sqrt{5}) = a^2 - 5b^2$. For nonzero rational a, b the latter is nonzero, since $\sqrt{5}$ is irrational. All operations and comparisons are rational.

The bundled `verify_exact.py` requires Python 3 and SymPy. Run `python3 verify_exact.py`. It checks normalization, positive A_k, B_k, both Gram determinants for each example, all eight strict normal inequalities for each example, nonnegative dual weights, and all six positivity conditions for each exclusion certificate. Its output is also included as `VERIFICATION.txt`. No solver, tolerance, or floating-point sign decision is used. A second bundled script, `verify_normal_identities.py`, checks the roots-of-unity identities and all forty normal-product identities for both modes and symbolic θ. The general equivalence and lower-product coverage are proved in the manuscript rather than inferred from finite tests.

Run `bash build.sh` with pdfLaTeX to reproduce this manuscript. The source archive includes the unchanged reference-style package used for the preceding reports, the manuscript, the build script, the exact verifier, and its output.

6.1 Expanded feasible certificates at $\theta = 1/8$

For compactness, write $z = \sqrt{5}$ and $\Delta_k = A_k B_k - c_k^2 - s_k^2 e_k^2$. Substituting (15) gives

$$\Delta_1 = \frac{489363679}{10^{12}} - \frac{1014049}{5000000000} z > 0,$$
$$\Delta_2 = \frac{-142775274757 + 65522176729 z}{8000000000000} > 0.$$

The eight strict normal inequalities are the signs of the following exact values:

$$P_1 = \frac{343747}{1600000} - \frac{19281221}{200000000} z < 0,$$
$$P_2 = \frac{343747}{1600000} - \frac{24721457}{200000000} z < 0,$$
$$P_3 = \frac{23747}{800000} - \frac{1360059}{100000000} z < 0,$$
$$P_4 = -\frac{147281}{1600000} + \frac{5200261}{200000000} z < 0,$$
$$H_1 = -\frac{571401}{6400000} + \frac{316819}{8000000} z < 0,$$
$$H_2 = \frac{-3809079 + 1699299 z}{12800000} < 0,$$
$$H_3 = -\frac{1756413}{6400000} + \frac{489629}{4000000} z < 0,$$
$$H_4 = -\frac{95779}{800000} + \frac{208013}{8000000} z < 0.$$

Each sign can be checked by the rational rule above. Together with positive A_k, B_k and $A_1 + A_2 = 1$, this proves feasibility directly, with no optimization step. Rotation invariance makes each displayed P_j or H_j occur at all five phases, proving all forty strict normal conditions of the actual indicator construction.

6.2 Expanded feasible certificates at $\theta = 1$

Substituting (16) gives

$$\Delta_1 = \frac{2215146143 - 930311001z}{125000000000} > 0,$$
$$\Delta_2 = -\frac{10682202001}{10^{12}} + \frac{49098049}{7812500000}z > 0.$$

The eight strict normal expressions are

$$P_1 = \frac{590297}{1400000} - \frac{2661817}{14000000}z < 0,$$
$$P_2 = \frac{590297}{1400000} - \frac{864189}{2800000}z < 0,$$
$$P_3 = \frac{90297}{700000} - \frac{207391}{3500000}z < 0,$$
$$P_4 = -\frac{1434467}{5600000} - \frac{2005363}{28000000}z < 0,$$
$$H_1 = \frac{19319}{800000} - \frac{49463}{4000000}z < 0,$$
$$H_2 = \frac{-448913 + 199507z}{800000} < 0,$$
$$H_3 = -\frac{214797}{800000} + \frac{474039}{4000000}z < 0,$$
$$H_4 = -\frac{486509}{800000} - \frac{286481}{4000000}z < 0.$$

Again the rational sign rule verifies each strict inequality. These two-mode witnesses prove nonconvergence of their full paired iterates, but do not establish any global convergence boundary, attraction property, or classification of other periods. The exact feasibility system is the classification supplied here; the two numerical parameter slices have certified rational brackets only.

References

[1] A. Chambolle and T. Pock, A first-order primal-dual algorithm for convex problems with applications to imaging, *Journal of Mathematical Imaging and Vision* **40** (2011), 120–145. doi:10.1007/s10851-010-0251-1.